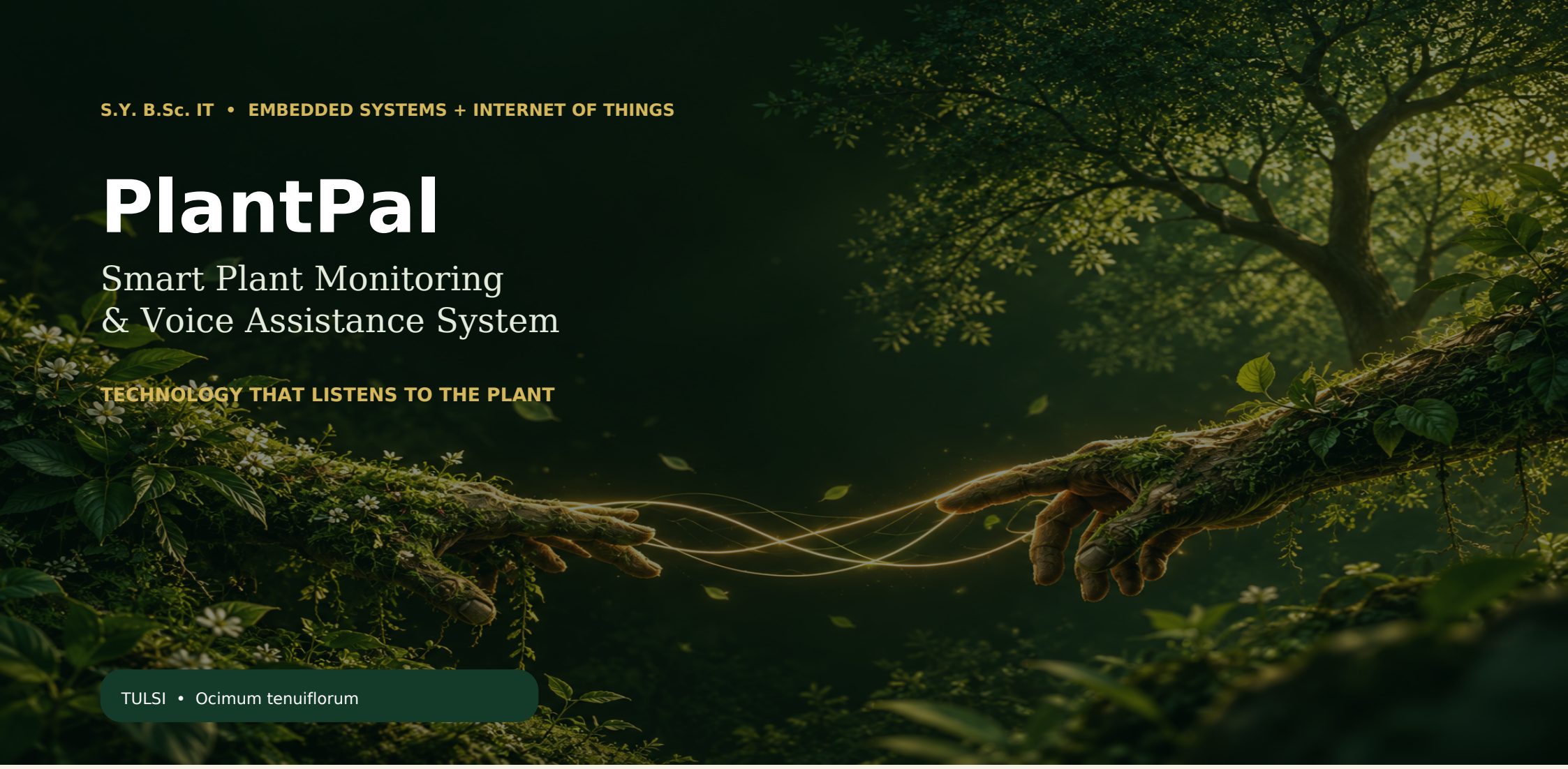
S.Y. B.Sc. IT • EMBEDDED SYSTEMS + INTERNET OF THINGS

PlantPal

Smart Plant Monitoring & Voice Assistance System

TECHNOLOGY THAT LISTENS TO THE PLANT



TULSI • Ocimum tenuiflorum

THE PROBLEM

Plant care often relies on occasional visual checks. Changes in soil moisture, light and surrounding conditions may go unnoticed.

PlantPal makes readings visible and communicates a simple condition summary to the caretaker.

OUR APPROACH

A living Tulsi plant is monitored by an ESP32 and low-cost sensors. The device interprets readings locally, then shares selected data with a web dashboard.

Local feedback is designed to remain useful when cloud connectivity is unavailable.

HOW IT WORKS

01 · SENSE — capacitive soil sensor, BH1750 light sensor, DHT11 temperature/humidity sensor and TTP223 touch input.

02 · PROCESS — ESP32 validates readings, classifies project-defined states and calculates a composite health score.

03 · RESPOND — 128×64 OLED, DFPlayer Mini + speaker, touch interaction and buzzer.

04 · CONNECT — Wi-Fi sends telemetry to a Flask REST API; SQLite stores readings/events and the browser dashboard presents them.

SYSTEM AT A GLANCE

Controller: ESP32-WROOM-32 / Dev Module

Inputs: Soil moisture · Light · Temperature · Humidity · Touch

Local outputs: OLED · 73 stored voice tracks · Speaker · Buzzer

Cloud: Wi-Fi · HTTPS/REST · Flask · SQLite · Web dashboard

Design principle: local-first monitoring; human-controlled watering.

SYSTEM FLOW

PLANT + SENSORS
Moisture • light
climate • touch

›

ESP32 EDGE
Read • validate
classify • score

›

LOCAL FEEDBACK
OLED • voice
buzzer

›

IOT / CLOUD
HTTPS • Flask
SQLite • dashboard

WHY IT IS USEFUL

- Combines multi-sensor readings with OLED, touch and voice feedback.
- Uses an event-driven 73-track voice library.

ENGINEERING BOUNDARIES

- Soil percentage needs calibration in the final soil and pot.
- Thresholds are project-defined, not laboratory plant-health measurements.

PROJECT TEAM

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K. E. S. Shroff College (Autonomous) • 2026–27